EXPERIMENT 4

Data Modelling and Form Design for a Business Application in Zoho Creator

Gym Membership Management System – forms, fields and lookup relationships

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1. Aim

To design the data model and forms of the Gym Membership Management System in Zoho Creator: six forms joined by lookup fields, using a wide range of basic field types, so that members, trainers, plans, classes, payments and attendance are stored in a structured way.

2. Procedure

Step 1: Create the application. An application named Gym_Membership_Management_System was created and six forms were added: Trainer_Registration, Membership_Plans, Member_Registration, Class_Schedule, Payment_Tracking and Attendance_Tracking.

Step 2: Add the basic fields. Fields were dragged from the Basic Fields palette: single line, number, decimal, phone, e-mail, address, date, time, drop-down, multi-line and check box fields.

Step 3: Apply field formats. Phone fields use the India country code (+91); date fields use dd-MMM-yyyy; time fields use HH:mm:ss; Plan Fee and Amount Paid use the format ##,##,###.##; Gender, Membership Status, Payment Method, Payment Status and Attendance Status are drop-downs with fixed choices.

Step 4: Build the data model. Lookup fields link the forms: Member_Registration → Membership_Plans, Class_Schedule → Trainer_Registration, Payment_Tracking → Member_Registration and Membership_Plans, Attendance_Tracking → Member_Registration and Class_Schedule.

Step 5: Test the forms. Each form was opened in live mode to confirm that every field, format hint and choice list appears as designed.

3. Output

3.1 Data model

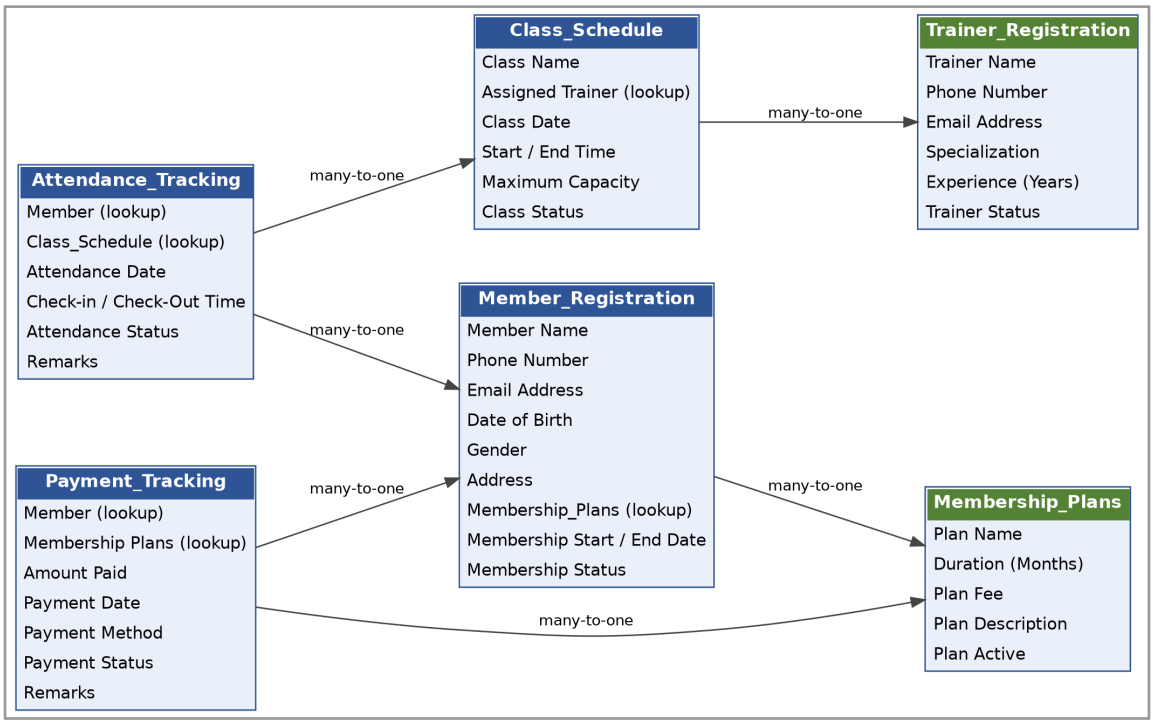


Figure 1: Entity–relationship (data model) diagram: six forms and six lookup relationships.

Trainer_Registration and Membership_Plans are master forms. Each lookup keeps one source of truth: a plan or trainer is entered once and then chosen from a drop-down everywhere else, instead of being retyped.

3.2 Master forms

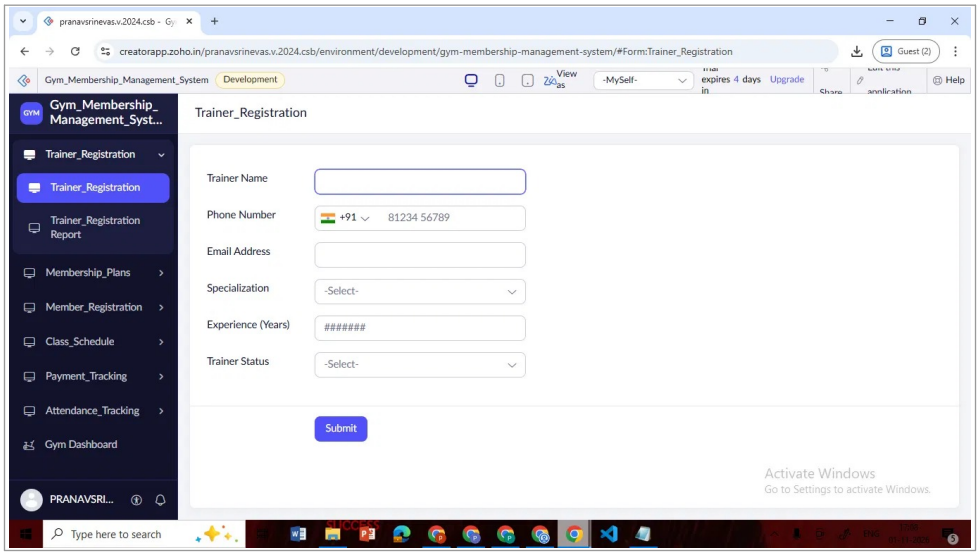


Figure 2: Trainer_Registration form: name, phone (+91), e-mail, specialization, experience and status.

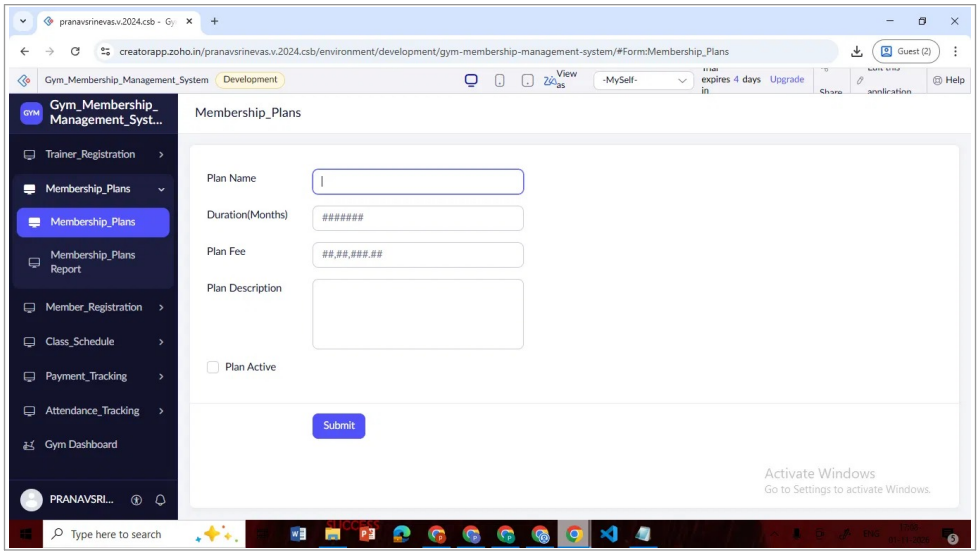


Figure 3: Membership_Plans form: plan name, duration (months), plan fee, description and Plan Active check box.

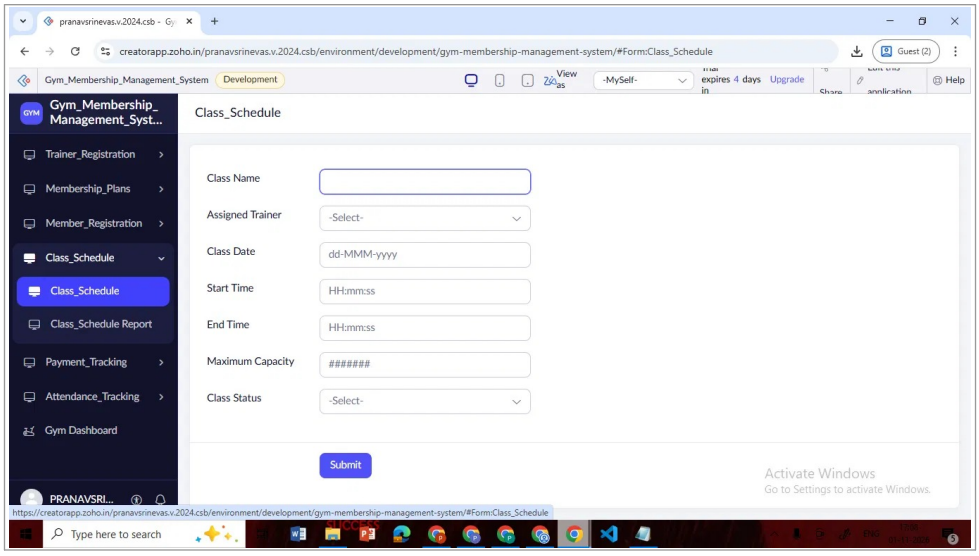


Figure 4: Class_Schedule form: class name, assigned trainer (lookup), date, start and end time, maximum capacity and status.

3.3 Member_Registration form

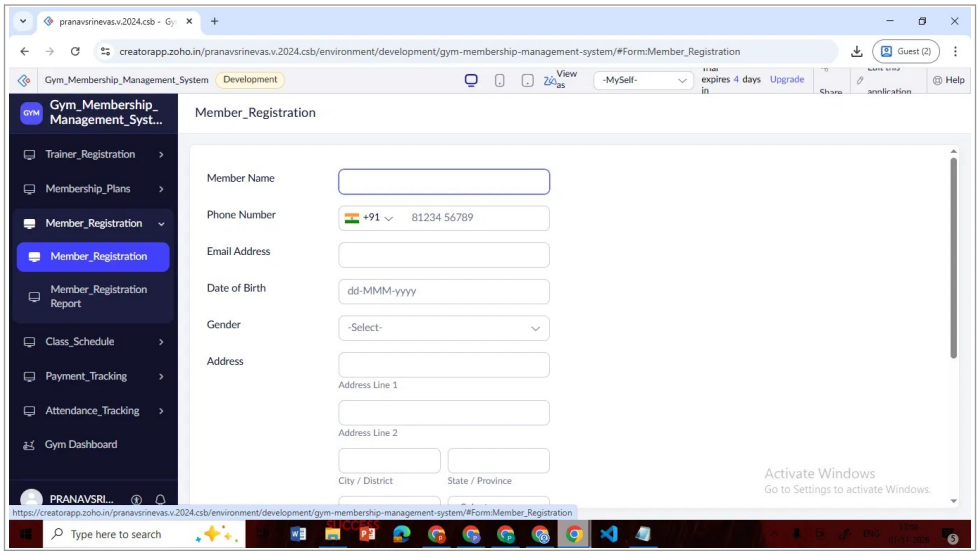


Figure 5: Member_Registration form: personal details, +91 phone, date of birth, gender and address.

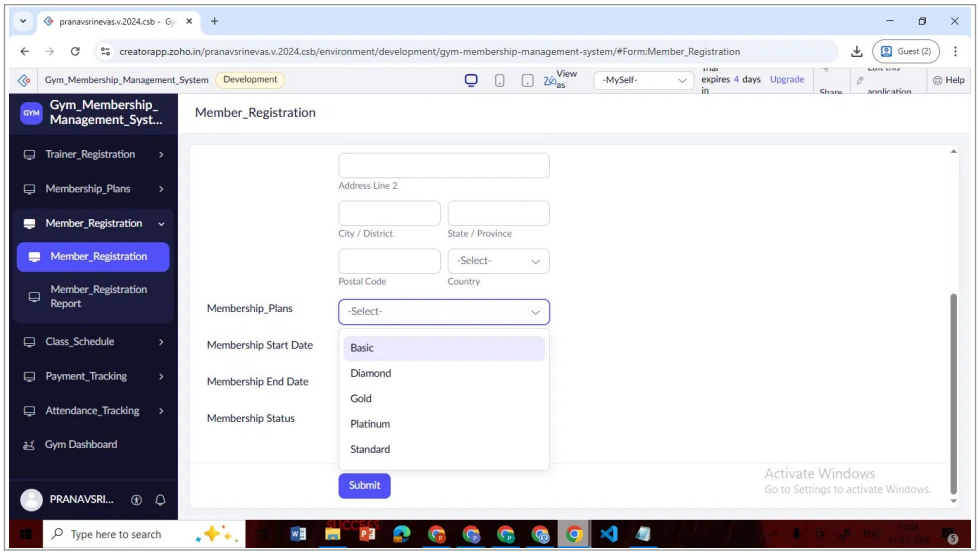


Figure 6: The Membership_Plans lookup lists Basic, Diamond, Gold, Platinum and Standard.

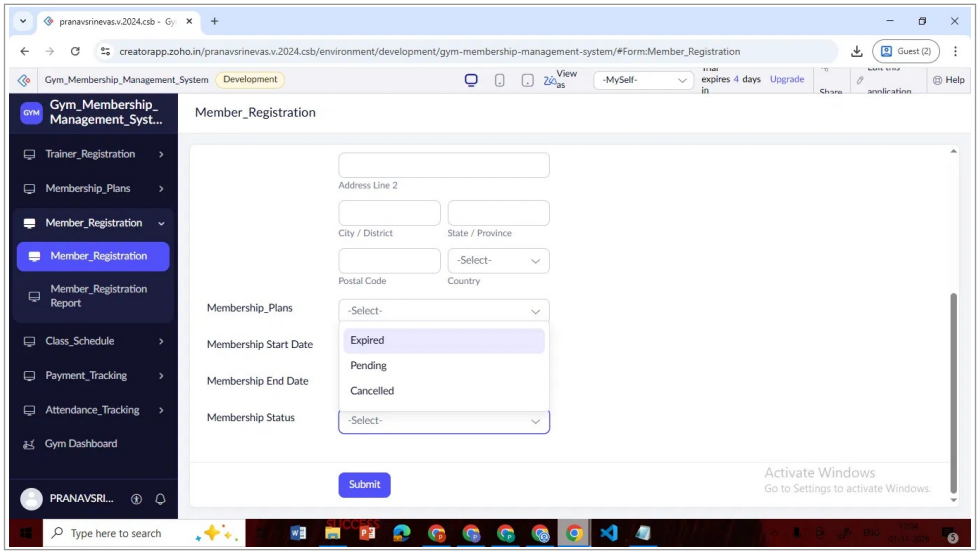


Figure 7: Membership Status drop-down choices: Expired, Pending and Cancelled.

3.4 Transaction forms

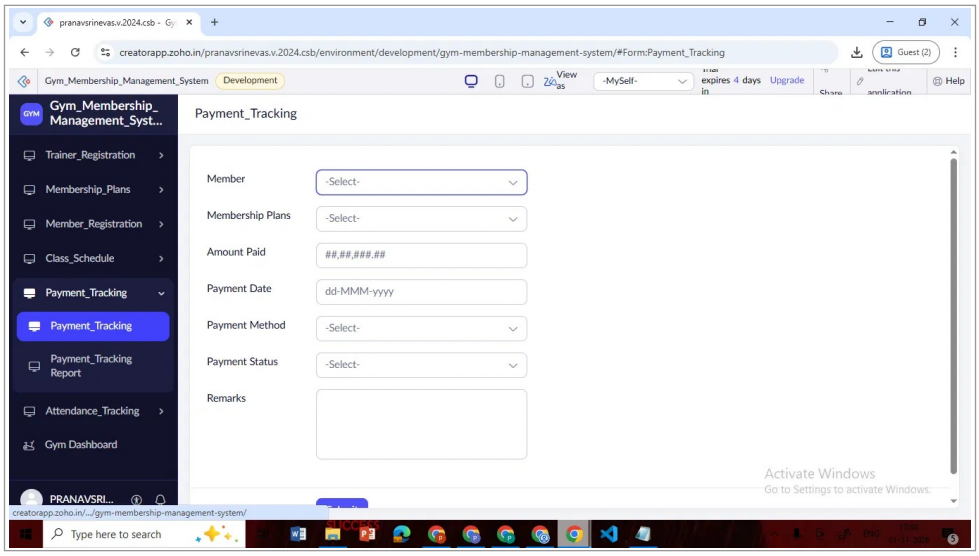


Figure 8: Payment_Tracking form with Member and Membership Plans lookups, amount, date, method, status and remarks.

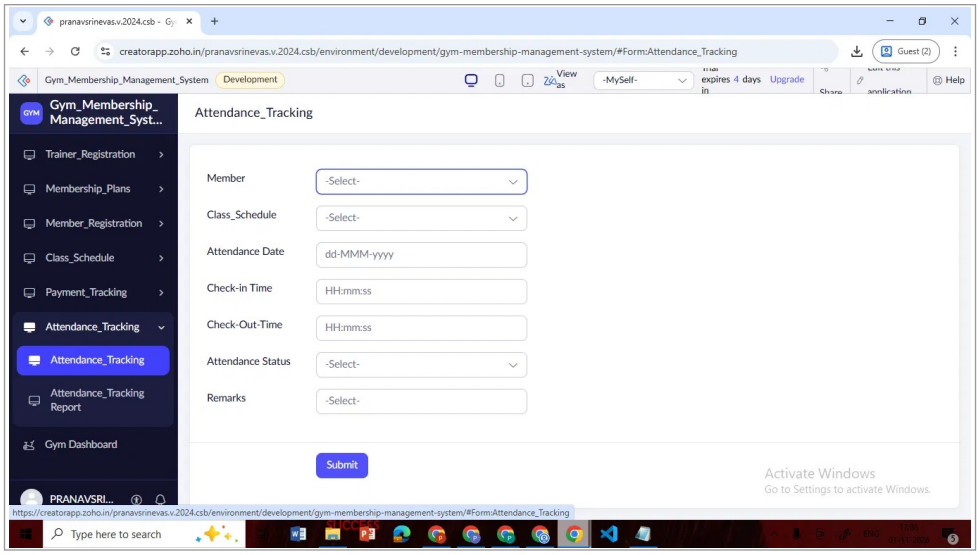


Figure 9: Attendance_Tracking form with Member and Class_Schedule lookups, date, check-in and check-out times and status.

3.5 Field summary

Form	Fields	Lookups
Trainer_Registration	Trainer Name, Phone, Email, Specialization, Experience (Years), Trainer Status	—
Membership_Plans	Plan Name, Duration(Months), Plan Fee, Plan Description, Plan Active	—
Member_Registration	Name, Phone, Email, Date of Birth, Gender, Address, Start and End Date, Membership Status	Membership_Plans
Class_Schedule	Class Name, Class Date, Start Time, End Time, Maximum Capacity, Class Status	Assigned Trainer
Payment_Tracking	Amount Paid, Payment Date, Payment Method, Payment Status, Remarks	Member, Membership Plans
Attendance_Tracking	Attendance Date, Check-in Time, Check-Out-Time, Attendance Status, Remarks	Member, Class_Schedule

4. Final Result

The data model of the gym application was built from six forms joined by six lookup fields, using phone, e-mail, address, date, time, number, decimal, drop-down, multi-line and check box fields. The formats and choice lists keep the data consistent, and the lookups avoid duplicated data. Custom validation scripts (such as age limits) were not part of this application; its only script is the welcome-mail workflow described in Experiment 6.